

CANON-SIGN 2.0RCT 1.4PUBLIC INPUTS

ONEBEHALFOF™ Receipt Verifier

ONEBEHALFOF™ verification of A⁵O appointment receipts: confirm who appointed the agent, on whose behalf it acted, and whether that appointment stood at the moment of execution.

The body, published keys, and dual signatures agree. This establishes integrity: any alteration to the receipt is detectable. It does not by itself establish authenticity — see Trust Anchor below.

Verify with ONEBEHALFOF™

See the trust boundary ↓



PUBLIC INPUTS
Receipt · public keys · appointment evidence

NEVER TRANSMITTED
Private keys · signer · decision logic

LIVE CONFORMANCE TEST

In-browser verification console

This is the live public verifier for the A⁵O Gate at theapertures.app. Every check runs in this browser from public evidence alone.

01
Authentic artifact

02
Alter principal

03
Swap published key

04
Revoke appointment

05
Reorder only

RECEIPT ARTIFACTASO-RCT-1.4

DECISION
ALLOW
F-STOP
f/1.2

RECEIPT
RCT-ASO-PUBLIC-DEMO-2026-07-10-0001

APPOINTER
did:a5o:appointer:agent-obo-inc

ONEBEHALFOF
did:a5o:principal:visionary-member-0710

AGENT
did:a5o:agent:iris-01

ACTION
book_private_aviation · TEB-NCE

RECEIPT HASH
62e03d2cee69c50b2f...3218ce9dac3b

INSPECT JSONDOWNLOAD ARTIFACT

PUBLIC VERDICT

ONEBEHALFOF™ verified

The body, published keys, and dual signatures agree. This establishes integrity: any alteration to the receipt is detectable. It does not by itself establish authenticity — see Trust Anchor below.

✓ RECEIPT BODY
ASO-RCT-1.4 structure
All 13 fields of the hashed body are present and declared-typed. The seal — receiptHash and both signatures — sits outside the hashed body by construction.

✓ RECEIPT BODY
Canonical field order
onebehalfof occupies position 7 in the deterministic body.

✓ RECEIPT BODY
Receipt hash
62e03d2cee69...cedac3b recomputed locally over the body.

✓ PUBLISHED KEYS
Published keyset digest
7afef9e892a...27eca42b matches the pinned public record.

✓ PUBLISHED KEYS
Receipt keys are published
Both embedded verification keys match active entries in the published keyset.

✓ DUAL SEAL
Ed25519 signature
Verified independently over the UTF-8 ASCII-hex receiptHash message.

✓ DUAL SEAL
ML-DSA-87 signature
FIPS 204 Category 3 signature verified over the same message.

✓ AUTHORITY EVIDENCE
Appointment reference
6508390ea2...3c5d9c is bound into the signed body. The hash commits to an appointment artifact; its contents are not disclosed and are not recoverable from this receipt.

✓ AUTHORITY EVIDENCE
Principal binding
Appointer, agent and onebehalfof resolve as three distinct, matching roles.

✓ AUTHORITY EVIDENCE
Scope and assertions
The receipt binds scope and asserts revocation epoch 42. Revocation state shown here is read from published evidence at the time of publication, not from a live operator query. Current state confirmation requires an external revocation oracle, which is not yet published. The signed scope includes credits<=150. No field in this receipt records the credit amount of the action, so conformance on that scope element is not witnessed by the receipt.

LOCAL VERIFICATION TIME856.0 ms

AUTHENTICITY BOUNDARY

Trust Anchor

This verifier runs entirely in your browser and never contacts the operator. Every input it consumes — receipt, published keyset, appointment evidence — is nonetheless served from this origin.

A passing verdict therefore establishes **integrity**: the artifact is internally consistent and any alteration is detectable. It does not by itself establish **authenticity**, which requires binding the published keyset to an anchor outside this operator's control.

KEYSET DIGEST7a9ef5fe592a...21eca42b

EXTERNAL ANCHORNot yet published

VERIFIER SOURCEInspect inline source

EVIDENCE, NOT ASSERTION

Inspect every public input

The seal stays portable. The verifier does not call the operator or replay the private gate.

Receipt

Published keyset

Appointment

Canonical body

```
receipt.json
{
  "receipt_id": "RCT-ASO-PUBLIC-DEMO-2026-07-10-0001",
  "schema_version": "ASO-RCT-1.4",
  "timestamp": "2026-07-10T14:08:00Z",
  "appointer": "did:a5o:appointer:agent-obo-inc",
  "appointment_ref": "sha256:d10b391b4b27feaa6cb089fe7d28327ae125862cd9e84ac600a9fefcf3e61c",
  "agent": "did:a5o:agent:iris-01",
  "onebehalfof": "did:a5o:principal:visionary-member-0710",
  "action": {
    "description": "book_private_aviation",
    "target": "TEB-NCE"
  },
  "scope": {
    "aviation.booking",
    "route:TEB-NCE",
    "credits<=150"
  },
  "decision": "ALLOW",
  "reason_code": "RRC-OK-VISIONARY-AVIATION",
  "revocation_as_of": 42,
  "fstop": "f/1.2",
  "receiptHash": "62e03d2cee69c50b2f67a130c1fd22e7843adda96c5cc4d49f5f3218ce9dac3b",
  "signatures": {
    "ed25519": {
      "alg": "Ed25519",
      "public_key": "d5fe8583e343e59612701cfeac6ab9e2b0c0026ae0b665e8c37cf5d5a78cae3",
      "signature": "27f1b07af73ff9806605a9026147e7ef1fbc072822107639caa01394e6368924413b7a2d6c677ce677cfde6a902fbc7517a1bf4a93b0be5c6aa8fd889d85109"
    },
    "ml_dsa87": {}
  }
}
```

Canonicalization — normative

ASO-RCT-1.4 canonical form. The hashed body is the receipt object with receiptHash and signatures removed, serialized as JSON with keys in the declared canonical order — not sorted — with separators , and ; UTF-8 encoded, with no whitespace, no prefix, no suffix and no trailing newline. For this receipt the preimage is 570 bytes.

receiptHash is the SHA-256 digest of that byte string, lowercase hexadecimal.

The signature message for both suites is the 64-character lowercase ASCII-hex encoding of that digest. The FIPS 204 context string is empty.

Note for implementers: sorted-key canonicalization (RFC 8785 / JCS) produces a different digest and will fail a valid receipt.

Verifier source — hashing and signature checks

```
const canonicalBody = JSON.stringify(body);
const calculatedHash = sha256(
  new TextEncoder().encode(canonicalBody)
);
const message = new TextEncoder().encode(calculatedHash);
ed25519.verify(signature, message, publicKey, { zip215: false });
ml_dsa87.verify(signature, message, publicKey);
```

The running page uses @noble/hashes, @noble/curves, and @noble/post-quantum for these checks. Scenario results are produced by the same verification path.

TRUST BOUNDARY

The proof travels. The secrets do not.

PUBLIC VERIFIER01

Receives evidence

Signed receipt artifact

Published public keyset

Appointment + revocation evidence

ONE-WAY RELEASE
→
public material only

OPERATOR BOUNDARY02

Retains control

Ed25519 + ML-DSA-87 private keys

Gate and policy engine

Signing ceremony + runtime state

PRIVATE MATERIAL NEVER SHIPPED

THE LOAD-BEARING DISTINCTION

“An unbound principal field is a label, not a proof.”

In RCT 1.4, onebehalfof sits inside the canonical receipt body. Change the principal and the hash changes. Change the hash and both signatures fail.